Binary cubic plane-quartic transfer screen

26 October 2026. Conductor T-5328. Frozen protocol, deterministic screen, raw result, independent replay and certificate. This record is a structural screen.

Result and novelty boundary

The fixed search drew 2,048 plane quartics over `k=GF(4)`, of which 55 were smooth with `#C(k)=5` and `#C(GF(16))=17`. Forty-eight of those had a Weil polynomial matching an elliptic curve over `K=GF(64)` in the complete 126-model source cohort. They occupied 12 distinct source-order classes. Eight of the 12 have no directly genus-three class-B source model in that cohort. This exhaustive absence concerns the 126 normalized elliptic models, not arbitrary covers or correspondences.

The strongest subgroup shapes in those eight are `#E(K)=58=2·29` and `62=2·31`. The direct binary GHS criterion gives genus four for their selected class-C source equations. Smooth plane quartics in their Weil-restriction isogeny classes were constructed and independently recounted. By Tate's finite-field theorem, equality of Weil polynomials implies an isogeny of abelian varieties. It does **not** supply an evaluated source-to-Jacobian map or establish that a chosen isogeny preserves the order-29 or order-31 group.

The LMFDB prior-work check changed the novelty verdict: **all eight isogeny classes already have catalogued Jacobian models, including plane quartics**. This search therefore does not establish a new eligible isogeny class or a new vulnerable curve. The open hypothesis is an *efficient explicit binary source-to-quartic transfer* with a complete decomposition attack and a favorable measured cost. Joux–Vitse already described characteristic-two hyperelliptic covers, decomposition, and isogeny walks; Tian's direct Weil-restriction quotient gives another prior approach to genus-three transfers. Neither a new attack method nor a below-rho cost is claimed here.

Exact candidates and prior-work checks

Let `w` satisfy `w^6+w^4+w^3+w+1=0` over `GF(2)`, and write integers in Sage's polynomial-basis encoding. Every source below is `E: y²+xy=x³+a*x²+b`. Let `a0` satisfy `a0²+a0+1=0` in `k=GF(4)`. The quartic equations are completely specified by the 15-element coefficient vectors in results.json: exponents are `(i,j,4-i-j)` for descending `i`, then descending `j`, beginning with `x⁴`.

#E	a,b	r, cofactor	N3	LMFDB class	catalogued Jacobians / hyperelliptic	rational point classes reach r?	distinct fully split lines
58	8,2	29,2	44	3.4.a_a_ah	10 / 4	yes	0
60	0,11	5,12	50	3.4.a_a_af	12 / 4	yes	0
62	8,8	31,2	56	3.4.a_a_ad	10 / 4	yes	0
68	0,8	17,4	74	3.4.a_a_d	16 / 4	yes	0
70	8,11	7,10	80	3.4.a_a_f	24 / 4	yes	0
72	0,2	3,24	86	3.4.a_a_h	18 / 4	no	0
76	0,10	19,4	98	3.4.a_a_l	20 / 4	yes	0
78	8,9	13,6	104	3.4.a_a_n	10 / 0	yes	0

The LMFDB `hyp_count=0` at order 78 says that its current catalog lists no hyperelliptic model; it is not a proof that none exists. The other seven classes already have catalogued hyperelliptic models as well as quartics. The database records were queried by exact degree, field size, full Weil polynomial and group order on 26 October 2026. Each class can be inspected at `https://www.lmfdb.org/Variety/Abelian/Fq/3/4/<label-tail>`; for example `3.4.a_a_ah` and `3.4.a_a_ad`. Targeted web searches on those labels and characteristic-two cubic cover attacks found no *evaluated source map* for these exact elliptic models. That limited search is not a proof of unpublished novelty.

For the two leading source groups, the independent verifier found:

source	G	Q=[7]G	prime-order check	quartic coefficient vector
a=8,b=2,#E=58	(41,55)	(24,54)	[29]G=[29]Q=O	[1,3,2,2,3,0,1,0,2,2,1,0,0,2,3]
a=8,b=8,#E=62	(9,36)	(46,53)	[31]G=[31]Q=O	[1,3,3,0,0,2,0,1,0,0,2,3,0,2,0]

These coordinates identify source subgroups and fixed target checks. They do not constitute an elliptic-to-quartic transfer. The verifier used Sage's independent projective rational-point enumeration for `N1,N2,N3`, recounted all 126 elliptic source groups, computed each quartic Jacobian's order with the Hess model, and checked whether rational point-difference classes reach the selected prime subgroup. All 12 witness orders passed the source-order, quartic-count and Jacobian-order checks. Eleven selected quartics passed the rational-point subgroup-reach check. For order 72, the five rational points in this selected quartic do not reach the order-three subgroup, making this particular factor base unsuitable.

GF(8)/GF(512) extension: stronger structural candidates

The separately frozen GF(8) protocol drew 1,024 quartics. Eleven were smooth with `N1=9`, `N2=65`, and a Weil polynomial matching one of the 1,022 exhaustively enumerated normalized source models; they covered ten group orders. The independent replay recounted every source group and all ten quartics over `GF(8)`, `GF(64)`, and `GF(512)`. It also computed the Jacobian group order and rational-point prime-subgroup reach for the four leading rows below. The frozen raw data and verification certificate contain every equation, coefficient vector and source model.

Here `K=GF(2)[w]/(w^9+w^4+1)`, `k=GF(2)[a0]/(a0^3+a0+1)`, integer coordinates use Sage's polynomial-basis encoding, and each source is `E: y²+xy=x³+x²+b`. The equations for each smooth plane quartic are the coefficient vectors in `results_q8.json` with the same fixed monomial order as above.

`#E(K)`	`b` `r`, cofactor	`G`, `Q=[7]G`	`N3` LMFDB class	catalogued models / hyperelliptic	distinct split lines
502	9 `251,2`	`(352,425)`, `(508,5)`	480 `3.8.a_a_al`	15 / 15	2
514	2 `257,2`	`(3,97)`, `(50,216)`	516 `3.8.a_a_b`	9 / 9	1
526	19 `263,2`	`(114,490)`, `(215,149)`	552 `3.8.a_a_n`	27 / 27	2
542	44 `271,2`	`(15,127)`, `(219,506)`	600 `3.8.a_a_bd`	15 / 15	2

The corresponding quartic coefficient vectors, in the monomial order above, are:

- Order 502: `[1,4,4,5,7,7,5,3,0,4,6,3,4,4,6]`.
- Order 514: `[1,1,5,4,6,5,7,4,3,0,3,2,6,2,6]`.
- Order 526: `[1,4,3,2,2,4,2,4,6,6,7,4,0,4,0]`.
- Order 542: `[1,1,7,6,2,5,1,2,6,5,0,0,3,7,4]`.

All source models at these four orders are class C in the complete cohort: none has the direct genus-three binary GHS trace condition. For each row, `[r]G=[r]Q=O`, the selected quartic is geometrically smooth, `#Jac(C)(k)=#E(K)`, and its nine rational point classes reach the order-`r` part. These checks establish a useful factor-base gate, not a transfer map.

The LMFDB exact-polynomial check identifies all four **isogeny classes** as previously catalogued, with only hyperelliptic equations in their currently stored `curves` arrays. Our plane quartic equations are explicit structural witnesses absent from those arrays; targeted searches on the four exact class labels and `plane quartic` returned no source identifying these exact equations. Neither catalog absence nor that bounded search proves model novelty. The potentially new result to pursue remains an evaluated binary source-to-quartic transfer and an improved complete attack on this cohort. The novelty labels for all four rows are: **new exact curve: unresolved**, **new eligible family: no**, and **new attack method: no**. The currently catalogued isogeny class does not establish a vulnerability. The four exact class records are `3.8.a_a_al`, `3.8.a_a_b`, `3.8.a_a_n`, and `3.8.a_a_bd`.

Attack-cost gate

For a source point `P`, the missing object is a computable geometric map `Φ : E(K) → Jac(C)(k)` (or a verified chain of such maps) with `Φ(G) ≠ 0` and `Φ(Q)=[7]Φ(G)` in the two leading cases. Matching zeta polynomials and equal group orders cannot replace that check. The next run must account for construction and evaluation of `Φ`, relation collection, linear algebra, target descent and independent scalar replay, then compare the complete cold cost with rho on the *same* `(G,Q)`. The current same-target GHS-versus-rho ratio is **unknown** for every row. The zero distinct fully split lines for the eight `GF(4)` quartics warns that a plain four-point line harvest there is inadequate. The four `GF(8)` quartics have one or two such lines, still far short of a demonstrated relation system; large-prime relations and target descent need separate measurements.

The repository's existing checked Lean subgroup lemmas are conditional on an additive nonzero map. This study does not instantiate their geometric hypotheses for these curves, and no new candidate-specific Lean proof is claimed.

Reproduction

From the repository root:

```
sage -python research/ghs_binary_cubic_isogeny_20261026/screen.py
sage -python research/ghs_binary_cubic_isogeny_20261026/verify.py
sage -python research/ghs_binary_cubic_isogeny_20261026/screen_q8.py
sage -python research/ghs_binary_cubic_isogeny_20261026/verify_q8.py
```

`results.json` binds the protocol and screen SHA-256 hashes; `verification.json` binds the raw result and verifier hashes. These files preserve the fixed-field run, including all 126 source models and the first quartic witness for each of 12 matched group orders. Elapsed runtime is a diagnostic only and is not an attack measurement. The corresponding `q8` files carry the full 1,022-source cohort and the first quartic witness for ten matched group orders.

Sources

- Joux–Vitse, cover and decomposition index calculus, especially §4.1 and Appendix A.2.
- Tian, cover attacks for elliptic curves with prime order, direct Weil-restriction isogeny construction.
- Diem–Thomé, genus-three plane-quartic index calculus.
- LMFDB, abelian varieties over finite fields.